CPSC 233 Assignments 01 Due Date: July 09th by 23:59

Overview:

This assignment covers the general programming concepts that you already know from your previous programming course. It will cover data types, variables, decision structures, loops, string and arrays/list/arraylist. You will write different methods to implement the functionalities. Below are the method that you need to write:

1. Write a static method ConvertToFenhenheit. It should convert Celsius temperature taken as an argument (type double), and convert this to Fahrenheit, and return the temperature (as a double). (3 marks)
2. Write a static method sumOddRange. It will take a start and end value as an integer. This method should return the sum of all odd numbers starting at start and ending at end. The last number should be excluded in the computation. For example, if start is 1 and end is 7 then the result should be 1 + 3 + 5 = 9. (3 marks)
3. Write a static method countChar. It will take a string and a character as an argument. The method should return an integer value. This value represents the number of times the character appear in the string. An upper case and lower case characters should be considered different characters. (3 marks)
4. Write a static method isPalindrome that takes two string as an argument. You should determine whether two strings are palindrome or not. A string is a palindrome, if it reads similar whether you read backward or forward. E.g. kayak, rotator are examples of palindrome. (3 marks)
5. Write a static method highestRatedMovie. This method will take the movieList and ratingList as argument. It should return a string, representing the highest rated movie. (4 marks)
6. Write a static method lowestRatedMovie. This method will take the movieList and ratingList as argument. It should return a string, representing the lowest rated movie. (4 marks)

Below are the further instructions for Q5 and Q6:

Assumption: No two movies has the same rating. The size of movieList and ratingList are same (you can fix it to be 5). Index in movieList corresponds to the index in ratingList. You can use ArrayList/Arrays for this question.

You will create one class for all of the above 6 questions. You can name BasicJava1 for the name of the class. In addition to writing above methods, you will also write main function that demonstrates the above functionalities. It means calling above methods from your main method. You should take the input in your main method.

It is important to note that your code should be able to run from the command line. Your TA should not install/delete any packages/module to run the program.

If you are facing any technical difficulty especially using Lab machines, send your email to Shokoofa zahra.ghods@ucalgary.ca. She is also added in the Discord server.

Collaboration

Discussing the assignment requirements with others is a reasonable thing to do, and an excellent way to learn. However, the work you hand-in must ultimately **be your work**. This is essential for you to benefit from the learning experience, and for the instructors and TAs to grade you fairly. Handing in work that is not your original work, but is represented as such, is plagiarism and academic misconduct. Penalties for academic misconduct are outlined in the university calendar.

Here are some tips to avoid plagiarism in your programming assignments.

1. Cite all sources of code that you hand-in that are not your original work. You can put the citation into comments in your program. For example, if you find and use code found on a web site, include a comment that says, for example:
   ```
   # the following code is from
   https://www.quackit.com/python/tutorial/python_hello_world.cfm.
   ```
   Use the complete URL so that the marker can check the source.
2. Citing sources avoids accusations of plagiarism and penalties for academic misconduct. However, you may still get a low grade if you submit code that is not primarily developed by yourself.
3. Discuss and share ideas with other programmers as much as you like, but make sure that when you write your code that it is your own. A good rule of thumb is to wait 20 minutes after talking with somebody before writing your code. If you find yourself exchanging code by electronic means, writing code while sitting and discussing with a fellow student, typing what you see on another person's console, then you can be sure that "your" code is not substantially your own, and your sources must then be cited to avoid plagiarism.
4. **Collaborative coding is strictly prohibited.** Your assignment submission must be strictly your code. Discussing anything beyond assignment requirements and ideas is a strictly forbidden form of collaboration. This includes sharing code, discussing code itself, or modeling code after another student's algorithm. **You cannot use (even with citation) another student's code.**
5. We will be looking for plagiarism in all code submissions, possibly using automated software designed for the task. For example, see Measures of Software Similarity (MOSS - https://theory.stanford.edu/~aiken/moss/).

Remember, if you are having trouble with an assignment, it is always better to go to your TA and/or instructor to get help, than it is to plagiarize.

Grading:

There are total 20 marks for this assignment. There will be five assignments in this course. All of the assignments have equal weight-age.